

Agahi (awareness) gives farmers and small traders selling into Kalimati, Kathmandu's main wholesale market, today's price, a price range 7 days to 3 months ahead, weather for their own district and a sell or keep tip, by plain SMS on any phone. Every reply fits in two SMS segments and works in English, Roman Nepali and Nepali. It runs on small models trained on local data, with **no large language model at runtime**.

13
crops with daily prices

91
crop x horizon forecast cells

2 MB
all 91 price models

0.66 ms
median CPU inference

99.4%
intent accuracy, held out

2 SMS
maximum reply length

How it works

A farmer texts the number, picks a district once, then uses a numbered menu or short free text such as *golbheda, alu 2 hapta* or *becham kauli 300kg*. Twilio forwards the SMS to a FastAPI webhook; the reply goes back in the same request (Twiml). A web simulator behaves exactly like SMS, and an admin dashboard shows conversations, forecasts, model evaluation, data health and SMS operations.

Pipeline

Stage	What happens
Understand	Text normalisation, Roman Nepali and Devanagari aliases, spell correction (rapidfuzz), TF-IDF word 1-2 + char 2-5 n-grams with logistic regression (runtime copy in NumPy, 1 MB). Slots: crop, horizon, quantity, location. Follow-ups such as <i>ani?</i> keep context.
Decide	Menu state machine; 0.75+ confidence acts, 0.45 to 0.75 asks <i>Did you mean</i> with the top two, below that shows the menu.
Answer	Price, forecast, weather, arrivals, sell or keep. Templates in 3 languages fitted to GSM-7 or UCS-2 segments, line breaks kept.
Send	Twilio (signature checked, idempotent), outbox with retries, delivery receipts, opt-out with STOP.

Price forecasting

Per crop and horizon (d7, d14, d21, d28, m1, m2, m3), six candidates compete: persistence, seasonal naive, mean reversion, Ridge, a small HistGradientBoosting model and a shrinkage blend. Features: price returns and z-scores, arrivals, rain and temperature lags. **Expanding-window walk-forward backtests** with an embargo of h days pick the model with the lowest error. Ranges come from **split-conformal calibration** on out-of-fold errors, each block calibrated only on earlier blocks.

Range	Use	Backtest coverage
50% (P25 to P75)	Shown to farmers as the likely range, rounded to Rs5, clipped to the crop's seasonal low and high	48.6% (target 50%)
80% (P10 to P90)	Admin and status gate	74.8% (target 80%)

An internal status gate (reliable, indicative, pattern only, unavailable) decides what is shown; the latest run has 16 reliable, 25 indicative and 50 pattern-only cells. Farmers always read *rough estimate*, and sell or keep advice only uses 7 or 14 day cells that pass the reliable gate.

Example: 7-day forecast reply (within 2 SMS)

Tomato price in 7 days
Likely Rs50 to Rs70 per kg
Likely about the same as today
Today: Rs55
Rough estimate.
1 Other time
2 Other crop
3 Sell or keep
0 Menu

Same reply in Roman Nepali
Golbheda: 7 din pachhi ko bhau
Sambhavit Rs50 dekhi Rs70 prati kg
Aaja jastai rahana sakchha
Aaja: Rs55
Moto anuman.

Live model output, tomato, origin 2026-10-03. The 80% range for the same cell is Rs37 to Rs94.

Admin dashboard

Overview, conversations, farmers, SMS operations, market and forecasts (50% and 80% ranges side by side), weather outlook, models and evaluation (backtest matrix, calibration, live accuracy ledger), data and training (ZIP upload, train, fetch new data), test console, alerts and broadcasts, quality queue. A phone-view simulator is built in for demos.

Weather

Days 1 to 7: Open-Meteo forecast for the farmer's district (free, no key, 3 hour cache, 5 second timeout). Weeks 2 to 4: Ridge models on weekly aggregates, kept only if they beat climatology by 3% in year-by-year tests (0 of 84 passed, so climatology is used and labelled). Months 1 to 3: day-of-year climatology nudged by the last 30 days. All outlooks are marked *estimated*; with no internet the app falls back to climatology.

Datasets and links

Dataset	Coverage	Source
Kalimati daily wholesale prices (min, max, avg per kg)	2023-08-01 to 2026-10-03, 13 crops, 24,163 anchor rows	kalimatimarket.gov.np/price
Kalimati daily arrivals	Same period, by commodity	kalimatimarket.gov.np/daily-arrivals
Daily weather: rain, Tmin, Tmax, wind, heat index	2015-01-01 to 2026-09-24, 6 districts + area mean, 29,995 rows	Google Earth Engine reanalysis export (earthengine.google.com)
Live 7-day forecast	Per district, at request time	open-meteo.com
NLU training set	9,240 labelled messages, 3 languages, plus 162 casual test messages	Written for the project (data/nlu_*.jsonl)

New Kalimati days are fetched automatically on the first request of each Nepal day and by a daily Vercel Cron call, appended once only, and today's estimates are refreshed with the trained models (no retraining).

Stack and deployment

Layer	Choice
Backend	Python, FastAPI, SQLite (WAL), Jinja2, vanilla JS, Chart.js
Models	scikit-learn for training only; runtime is NumPy, pandas, joblib (about 200 MB, fits Vercel's 250 MB limit)
SMS	Twilio inbound webhook, Twiml reply, REST outbound, status callbacks
Hosting	Vercel serverless; seed DB and models in deploy_assets/ (16 MB), cold start copies them to /tmp
Quality	93 automated tests, 33 smoke checks, 40 scenario conversations, every template checked against the 2-segment limit

Honest limits

Prices are Kalimati wholesale, not farm-gate. Just over three years of history, so month-ahead ranges are seasonal estimates and most cells are pattern only. Arrivals include imports. Nepali and Roman Nepali text needs review by a native speaker. Fetching on Vercel resets on a cold start until assets are redeployed.

Run it

Step	Command or action
Local	Unzip, then run .bat (Windows) or ./run.sh; open localhost:8000 and /admin. First run trains in about 3 minutes
Deploy	Push to GitHub, import in Vercel, set ADMIN_PASSWORD, SECRET_KEY, CRON_SECRET and the Twilio keys
SMS	Point the Twilio number's incoming webhook at /api/sms/inbound (HTTP POST)